Conditional Probability and Its Applications

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1 Introduction

Ok, so here's the thing.... Have you ever played a playlist on Spotify on shuffle? Most probably, yes; after all, Spotify is one of the most widely used music streaming services. Most people actually think it's the algorithm pushing songs to us and ruining our music, but sometimes what we perceive as a problem is simply randomness appearing in clumps. Even more interestingly, Spotify has adjusted its shuffle algorithm to make the results feel more random to listeners.

So why does this happen in the first place? It's quite simple: human intuition often expects random events to look evenly spread out. We may think a shuffle button makes each song selection completely independent, but a playlist shuffled without replacement is affected by which songs have already been played. More generally, in many real-world situations, the probability of what happens next is shaped by the context of what has already happened, especially when events are dependent. This is the idea behind conditional probability.

2 Mathematical Representation

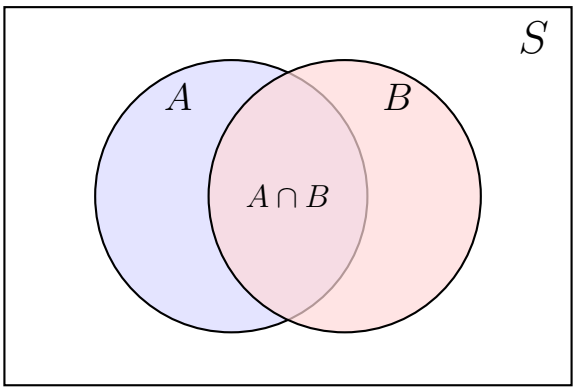
To understand how we calculate probabilities in real-world situations, we use the notation $P(A|B)$. The vertical line represents the phrase "given that," which asks: what is the probability of Event A happening, given that Event B has already occurred?

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{for } P(B) > 0 \tag{1}$$

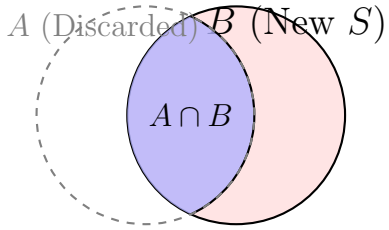
In simpler words, we take the probability of both events overlapping ($A \cap B$) and divide it by the probability of the condition, $P(B)$. This division reduces the entire universe of possibilities (the sample space) down to just the scenarios where Event B is valid.

3 Visualizing Conditional Probability

To picture this mathematically, imagine the entire universe of possible outcomes as a box. Inside this box are two overlapping circles representing Event A and Event B.



After: The Shrunken Universe (Given B)



When we calculate $P(A|B)$, we assume that Event B has already occurred. In the above diagram, we are going to completely ignore everything in the rectangle outside circle B. Circle B becomes our new universe. We then look exclusively at the overlapping part of A and B, $A \cap B$ (the numerator), to see where A exists inside this new reality.

4 Properties of Conditional Probability

Before we move to the applications, we have to understand why and in which scenarios conditional probability is going to work.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $B \in \mathcal{F}$ be an event such that $P(B) > 0$. The conditional probability measure $P(\cdot \mid B)$ preserves the foundational axioms of probability within the restricted sample space:

1. **Axiom 1: Non-Negativity**

$$P(A \mid B) \geq 0 \quad \forall\, A \in \mathcal{F}$$

The conditional probability of any event in the new sample space B is always going to be non-negative.

2. **Axiom 2: Normalisation (Certainty)**

$$P(\Omega \mid B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

Similarly,

$$P(B \mid B) = \frac{P(B \cap B)}{P(B)} = 1$$

The new restricted sample space B becomes the certain event because we are already given that B has occurred.

3. **Axiom 3: Countable Additivity**

   For any mutually exclusive events $A_1, A_2, \ldots$ (where $A_i \cap A_j = \emptyset$ for $i \neq j$):

$$P\left(\bigcup_{i=1}^{\infty} A_i \,\middle|\, B\right) = \sum_{i=1}^{\infty} P(A_i \mid B)$$

Connection to Independence

Conditional probability is also closely related to the concept of independence.

Two events A and B are said to be independent if the occurrence of Event B does not change the probability of Event A.

Mathematically, this can be represented as:

$$P(A \mid B) = P(A)$$

provided that $P(B) > 0$.

Using the conditional probability formula, this is equivalent to:

$$P(A \cap B) = P(A)P(B)$$

This shows how conditional probability helps us understand the difference between independent and dependent events.

5 Application of Conditional Probability

For our application, we are going to take the weather forecasting example to show how conditional probability can be used to calculate the likelihood of rain based on humidity. Let's consider the following scenario.

5.1 We are given:

Historical weather data for a city in November reveals the following probabilities:

- The probability of a day being highly humid is $P(\text{Humidity}) = 0.40$.
- The probability of a day having both rain and high humidity is $P(\text{Rain} \cap \text{Humidity}) = 0.20$.

Using Conditional Probability

Here, Humidity is going to work as B because we are calculating the probability of rain given that the day is already highly humid. This means that $P(\text{Humidity})$ represents our restricted sample space. High humidity does not guarantee rain, but it gives us a condition under which we can calculate the probability of rain.

Now, to find the probability that it will rain given that it is already highly humid, we use the conditional probability formula:

$$P(\text{Rain} \mid \text{Humidity}) = \frac{P(\text{Rain} \cap \text{Humidity})}{P(\text{Humidity})}$$

Substituting the known values into the equation:

$$\begin{aligned} P(\text{Rain} \mid \text{Humidity}) &= \frac{0.20}{0.40} \\ &= 0.50 \text{ or } \mathbf{50}\% \end{aligned}$$

Conclusion

This shows us how conditional probability can help meteorologists in weather forecasting. Based on the given historical data, for this specific question, it indicates that there is a **50% chance of rain** on a day that is already highly humid.

However, in an actual weather forecast, meteorologists consider several other factors as well.

Wrapping Up!

Conditional probability is not just a concept that exists in theory, but it also provides us with real-world applications, from weather forecasting to processing complex data structures in data science algorithms. Its real-world applications make it an important bridge between theoretical probability and practical situations.

Mastering this transition—from pure mathematical axioms to real-world applications—is essential for understanding more complex theories like Bayes' Theorem, Independence vs. Dependence, and the Law of Total Probability, which can be used to create real-time predictive models.